

Algebra: Solving Logarithmic Expressions

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DIRECTIONS

Evaluate each log expression or solve for x in the logarithmic equation.

1 Evaluate:

$$\log_9 3$$

Answer: _____

2 Solve for x:

$$\log_2 x = -3$$

Answer: _____

3 Solve for x:

$$\log_4 x = -1$$

Answer: _____

4 Solve for x:

$$\log_2 x = 5$$

Answer: _____

5 Solve for x:

$$\log_5 x = 3$$

Answer: _____

6 Evaluate:

$$\ln 1$$

Answer: _____

7 Evaluate:

$$\log_{10} 0.001$$

Answer: _____

8 Evaluate:

$$\log_4 2$$

Answer: _____

9 Evaluate:

$$\log_5 25$$

Answer: _____

10 Solve for x:

$$\log x = 2$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Convert to exponential form to solve: $\log_b(x)=y$ means $b^y=x$. Check for extraneous solutions.

1 Evaluate:

$$= \frac{1}{2}$$
$$\log_9 3$$

2 Solve for x:

$$= x = \frac{1}{8}$$
$$\log_2 x = -3$$

3 Solve for x:

$$= x = \frac{1}{4}$$
$$\log_4 x = -1$$

4 Solve for x:

$$= x = 32$$
$$\log_2 x = 5$$

5 Solve for x:

$$= x = 125$$
$$\log_5 x = 3$$

6 Evaluate:

$$= 0$$
$$\ln 1$$

7 Evaluate:

$$= -3$$
$$\log_{10} 0.001$$

8 Evaluate:

$$= \frac{1}{2}$$
$$\log_4 2$$

9 Evaluate:

$$= 2$$
$$\log_5 25$$

10 Solve for x:

$$= x = 100$$
$$\log x = 2$$



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Page 2